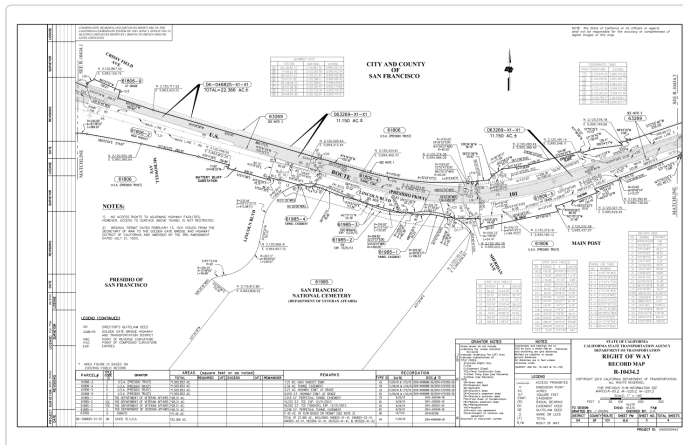


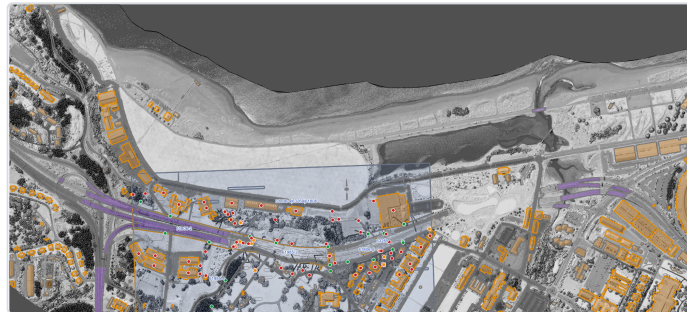
SWYFT Record Twin — one sheet, end to end

A right-of-way record map goes in. A georeferenced object record comes out: every line, curve, parcel and control point the sheet prints, each carrying what the sheet says, what the sheet draws, whether the two agree, and a citation back to the spot on the sheet. Anything the geometry cannot settle goes to a surveyor with a picture and a reason. Nothing is guessed.

1. What goes in



The record map — Caltrans right-of-way sheet R-10434.2 (Presidio Parkway, Route 101). A vector PDF: every character is a stroked path, every line is geometry. It carries three kinds of record: the drawing (lines, curves, parcel boundaries), the printed values beside them (bearings, distances, curve data), and the tables (line table, curve tables, coordinate table, parcel table).



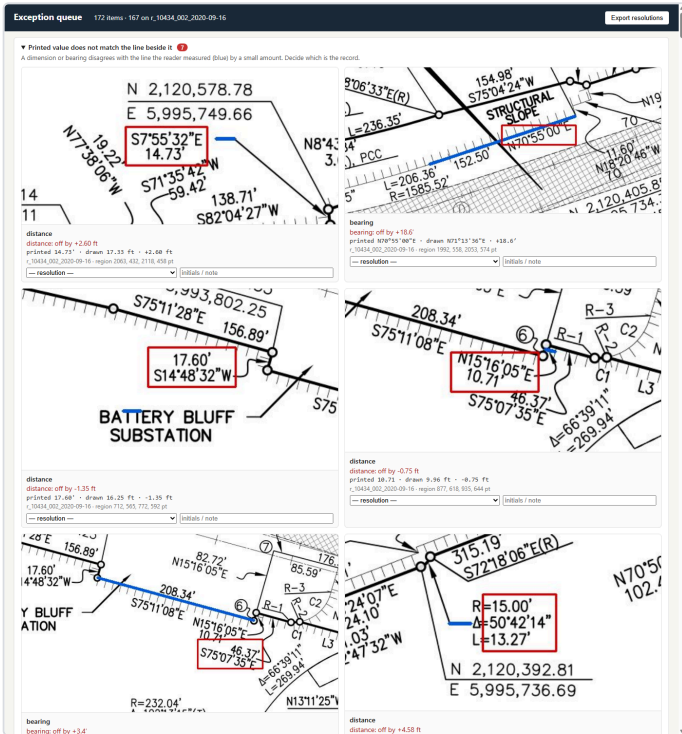
The LiDAR — a 2 km × 0.9 km airborne tile over the same ground, flown in 2025, shown here as a hillshade with the sheet's objects already placed on it. The sheet is on the 1991 epoch of the datum; the LiDAR on the 2010 epoch. The 0.67 m difference is applied before anything is compared.

2. What comes out

An **object record** (objects .geojson, 3,706 objects). Each object has: kind, the printed value, the measured value, the residual, the rule that measured it, a status, and the sheet region it came from.

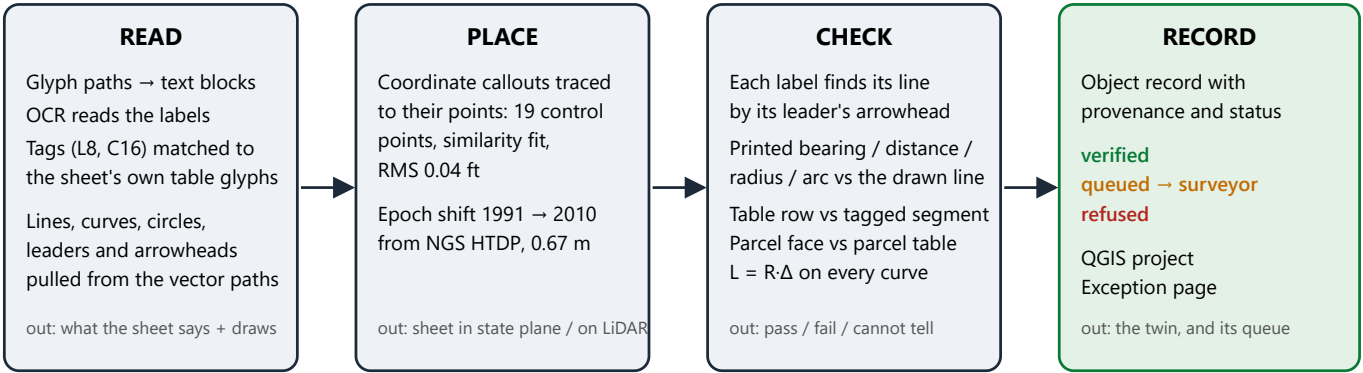
Status	Meaning	Objects
verified	printed value matches the drawing within tolerance	60
queued	geometry could not settle it; a surveyor decides, from a crop and a reason	220
refused	not checkable on this sheet (runs to a matchline, a radial, a total)	64
record	placed and cited, no check applies (plain linework, LiDAR features)	3,362

Three ways to look at it: a QGIS project with every layer styled and a hover tip per object; an offline exception page, one card per queued item; and a numbers file the slides quote from.



The exception page. Each card: a crop of the sheet (red = the label, blue = the line that was measured), printed vs drawn, the reason, a citation, and a resolution field the surveyor fills in.

3. What it is actually doing



LiDAR runs beside this: ground, pavement, deck and building footprints are extracted by rule and screened against the placed parcels.

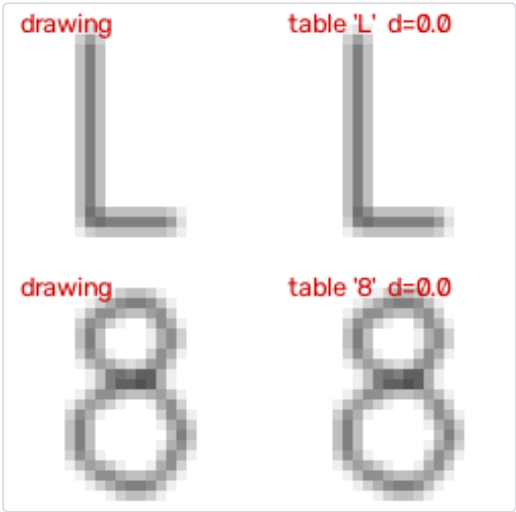
The principle. The record is the numbers. The drawing is checked against them, never the other way round, and no reading is trusted because a reader was confident — only because the geometry agrees. When two lines could be the one a label means, or a tag has a character the matcher cannot settle, the item is queued, not guessed.

4. One tag, followed through

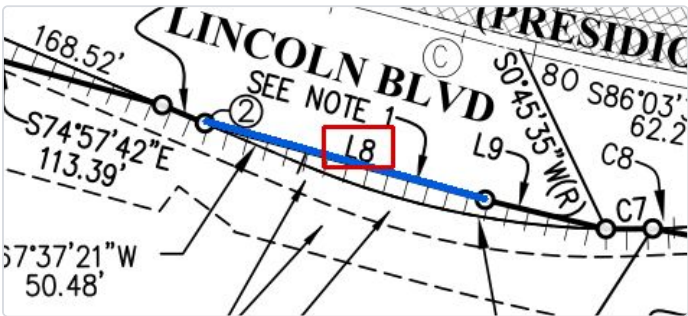
Line **L8** on this sheet. Four steps, each one a picture.

LINE DATA TABLE		
NO.	BEARING	DISTANCE
L1	N18°06'50"E(R)	1.86'
L2	N14°18'16"E	3.76'
L3	S75°11'28"E	96.27'
L4	S75°12'02"E	48.55'
L5	N72°08'15"E	19.34'
L6	S70°11'40"E	6.29'
L7	S74°42'15"E	39.52'
L8	S75°11'28"E	139.90'
L9	S76°56'20"E	59.89'

The record. The line table says L8 bears S75°11'28"E for 139.90 ft. Table values are read once and held as the record.



Reading the tag without OCR. Every character on the sheet is a stroked path. The table's own characters, whose values are known, are the alphabet; each character on the drawing is matched against them at 36 rotations. The "L" and "8" beside the line match the table's L and 8 exactly (distance 0.0).



Which line. The tag (red box) has a leader ending in an arrowhead; the arrowhead touches one line (blue). That line is measured through the georeferencing fit: 139.90 ft, azimuth S75°12'00"E.

The verdict.

check	printed	drawn	difference	result
distance	139.90 ft	139.90 ft	-0.00 ft	pass
bearing	S75°11'28"E	S75°12'00"E	0.5'	pass

Tolerances: 0.30 ft plus 0.05 % on distance; 3 arc-minutes on bearing, widened on short lines where a tenth of a foot sideways is a degree. The object record for L8 carries both rows, the rule, and the sheet region of the tag.

The same path runs for curves: radius from a circle fit, arc length from the drawn run. On this sheet the R = 1470 ft curve between two vertex circles measures 573.95 ft drawn against 573.93 ft printed. Where the record splits that run into arcs whose boundary is not drawn (C16 and C15 share it), the run is checked against the sum and labelled so.

7. The numbers, from the run

Stage	Result on R-10434.2
Georeferencing	19 of 19 control points used, RMS 0.042 ft. Caltrans' own georeferenced package of the same sheet is raster-limited to about 0.6–1.1 ft.
Labels on the drawing	distance 25 of 46 pass, bearing 12 of 20, arc length 3 of 13. Where a label passes it agrees to 0.03 ft median, 0.4' median bearing. 61 items queued: 8 genuine disagreements, 31 where the measured line is not the labelled one, 18 with no line found, 4 leaving the sheet.
Table tags	36 of 44 table rows found on the drawing, 7 partial reads queued, none misread. Bearing 9 of 11 pass, distance 8 of 10, radius 5 of 7, arc length 6 of 10.
Parcels	27 faces, 6 named. None within 1 % of the parcel table: the figures leave the sheet at matchlines and the easement strips do not close from linework alone.
LiDAR	32 pavement, 11 viaduct deck and 281 building polygons by rule; 175 buildings pass a roof-flatness test. 73 building-class clusters inside parcel faces, 9 with a building verdict.
Run	One command, 15 steps, 91 s, offline. Ends by writing the QGIS project, the figures and this table's source file.

8. What it does not do yet

Closure. Parcels are faces of the drawn linework, not traverses walked from the record; the tunnel easements on this sheet are drawn as dashed strips with inline values and do not close. **Scans.** A 1969 scanned sheet was tried: the readers agree on 1–2 of 87 coordinate boxes and no credible fit results; it sits in the queue as a sheet. **Drafter styles.** Leaders, tags and ticks are recognised by their geometry on this drafter's sheets; another drafter's conventions need their own adapter, which the queue will expose on the first sheet.